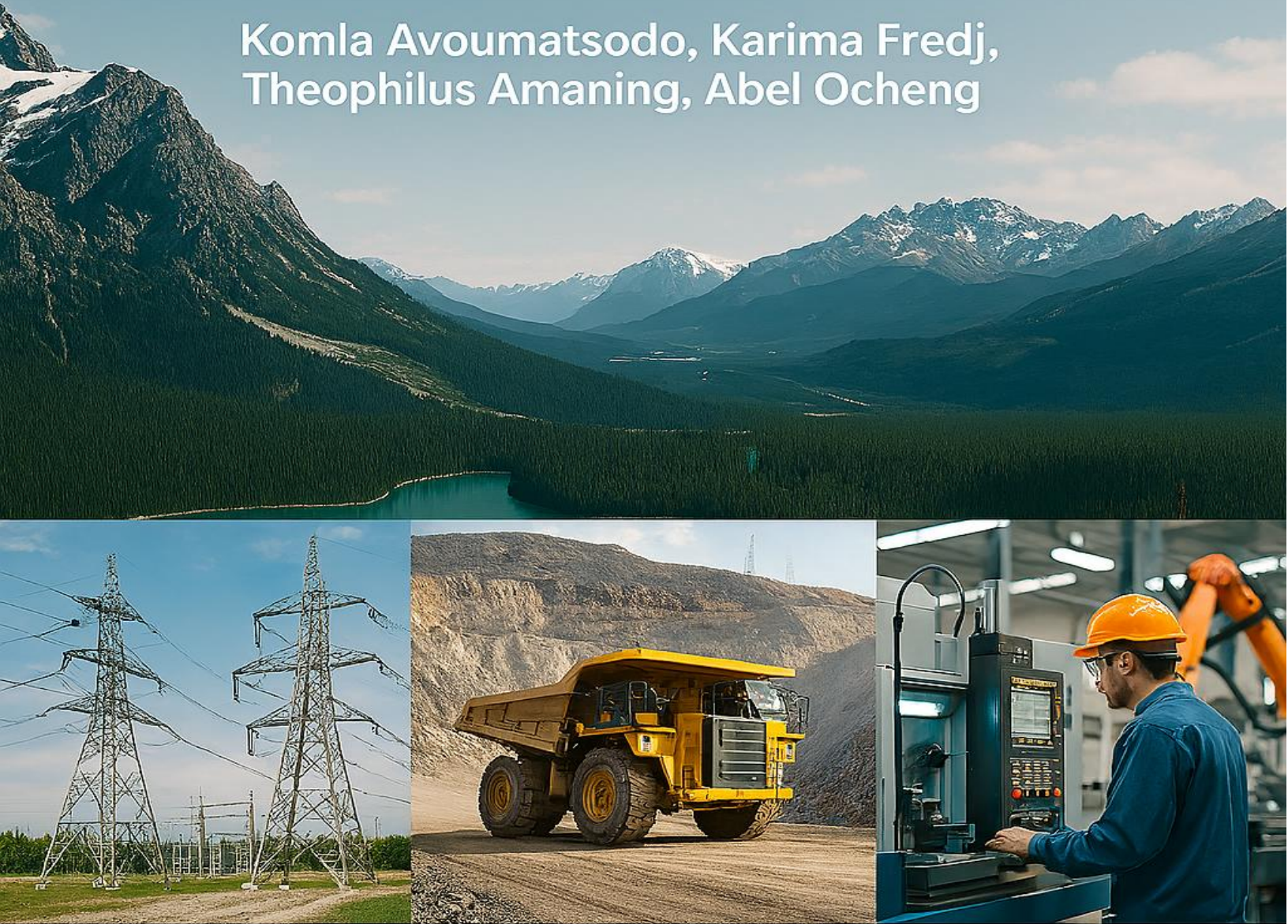


The Impact of BRITISH COLUMBIA'S PROVINCIAL SALES TAX ON INDUSTRIAL INVESTMENT

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The Impact of British Columbia's Provincial Sales Tax on Industrial Investment

A Policy Report

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List of Abbreviations

BC British Columbia

DiD Difference-in-Differences

GST Goods and Services Tax

GVBOT Greater Vancouver Board of Trade

HST Harmonized Sales Tax

M&E Machinery and Equipment

METR Marginal Effective Tax Rate

NAICS North American Industry Classification System

PST Provincial Sales Tax

UNBC University of Northern British Columbia

Executive Summary

Context. British Columbia (BC) has undergone two major shifts in its provincial sales tax regime since 2010. In July 2010, the province replaced its 7% Provincial Sales Tax (PST) with a fully creditable 12% Harmonized Sales Tax (HST), effectively eliminating the tax burden on business capital purchases. Following a referendum, BC reverted to the PST in April 2013. Unlike the HST, the PST is non-creditable: firms cannot recover it on purchases of machinery and equipment (M&E), raising the effective cost of capital by approximately 7 percentage points. This report quantifies the consequences of that policy reversal for industrial investment across BC.

What we find. Using a difference-in-differences framework that compares investment trends in BC to those in the nine other Canadian provinces over 2001–2024, we find that the 2013 PST reintroduction reduced M&E investment per worker in BC by approximately 6 per cent on average. The effect is larger in capital-intensive sectors and smaller in labour-intensive ones:

- Mining and Utilities experienced the steepest declines (–27.9% and –25.2% respectively relative to the pre-reform trend), consistent with their heavy reliance on taxable machinery.
- Construction recorded a moderate reduction (–6.1%).
- Services were essentially unaffected, reflecting their lower dependence on M&E investment.
- Within Manufacturing, traditional subindustries (food and beverage, wood products, paper, textiles) were more adversely affected than advanced ones (chemicals, electrical equipment, transportation equipment), consistent with differences in technological intensity and capital reliance.

These patterns are consistent with the PST operating through the *user cost of capital*: by raising the effective price of machinery and equipment, the tax discourages capital deepening in sectors most reliant on such assets.

Policy implications. The evidence points to five actionable recommendations for policymakers:

1. *Expand input tax credits or exempt capital inputs.* Broadening exemptions for M&E purchases from the PST would reduce the marginal effective tax rate on new investment, especially in capital-intensive industries.
2. *Simplify PST compliance rules.* The current exemption structure is complex and usage-dependent, creating uneven effective burdens and compliance costs that disproportionately burden smaller firms.
3. *Prioritise capital-intensive sectors in any reform.* Mining, Utilities, and traditional Manufacturing bear the heaviest distortions. Targeted relief in these sectors offers the largest potential gains.
4. *Evaluate a return to a value-added tax framework.* Harmonising BC's sales tax with the federal GST would eliminate cascading tax effects on capital inputs and align the province with the tax

treatment prevailing in Ontario, Quebec, and Atlantic Canada.

5. *Invest in sector-level tax burden panel data.* Detailed data on effective tax rates by sector, obtainable from the BC Ministry of Finance, would allow more precise measurement of the PST's incidence and strengthen the evidence base for future reform.

Bottom line: The 2013 reversion to the PST likely offset investment gains achieved under the HST. A reform that broadens input tax credits or reharmonises BC's sales tax with the federal GST could meaningfully improve the province's investment climate, capital deepening, labour productivity, and long-run economic growth.

1 Introduction

In April 2013, British Columbia reinstated its Provincial Sales Tax (PST) on business capital inputs, reversing a brief but economically significant experiment with a fully creditable Harmonized Sales Tax (HST). Unlike value-added taxes, the PST is non-creditable: firms pay it on purchases of machinery and equipment (M&E) and cannot recover the amount paid, directly raising the user cost of capital by approximately 7 percentage points relative to the HST period (Hall & Jorgenson 1967, Lammam 2016). The question this report addresses is whether that increase in capital costs translated into a measurable suppression of M&E investment in BC—and, importantly, whether the burden fell disproportionately on sectors most reliant on taxable capital.

The reform generates a compelling natural experiment. The PST reintroduction was province-wide, abrupt, and, crucially, triggered by a 2011 democratic referendum on the distributional consequences of the HST for consumers, not by investment conditions. This makes reverse causality implausible: deteriorating investment did not cause the reform. The other nine Canadian provinces, none of which experienced a comparable change in sales tax treatment of capital inputs, serve as the control group. This setup supports a difference-in-differences (DiD) identification strategy: we compare BC's sectoral investment trajectory before and after April 2013 to the contemporaneous evolution in control provinces, netting out both aggregate macroeconomic trends and permanent differences in provincial investment climates.

We find that the PST reintroduction reduced M&E investment per worker in BC by approximately 6 per cent on average—an estimate that is stable across four progressively richer control specifications and robust to the choice of estimation sample. The aggregate figure, however, conceals sharp and theoretically coherent heterogeneity. Mining investment fell by 27.9 per cent and Utilities by 25.2 per cent, while Services were essentially unaffected. Within manufacturing, traditional subindustries—food and beverage, wood products, paper, and textiles—were significantly more adversely affected than advanced ones: the PST imposed an additional 8.9 log-point reduction in investment per worker in traditional relative to advanced manufacturing, a differential that is statistically significant across all specifications. These patterns align precisely with the predictions of the user cost channel: the burden falls hardest on sectors most dependent on taxable capital equipment and least able to substitute away from it or absorb higher costs through margins.

The report makes four contributions. First, we provide rigorous causal evidence on the investment effects of a non-creditable retail sales tax, exploiting a within-jurisdiction policy reversal that cleanly separates the tax effect from confounding structural trends. Second, we document heterogeneous responses across six broad sectors, establishing that capital intensity is the primary mediator of the PST's burden. Third, we distinguish within manufacturing between technologically advanced and traditional subindustries, revealing how capital-skill complementarity and margin structure shape the investment response. Fourth, we translate these findings into five concrete reform recommendations calibrated to the magnitude and sectoral distribution of the distortion.

Section 2 situates the 2013 reform within BC's longer fiscal history and develops the user cost framework that underlies our empirical predictions. Section 3 describes the data and documents the descriptive investment patterns that motivate the formal analysis. Section 4 presents the econometric evidence, moving from average effects to sectoral and within-manufacturing heterogeneity. Section 5 develops the policy implications and recommendations. Section 6 concludes. Appendix A provides full technical details of the econometric design.

2 Policy Background: BC's Sales Tax History

Understanding the investment effects of the 2013 PST reintroduction requires more than a description of the reform itself: it demands situating that reform within BC's longer fiscal history and a clear theoretical account of how tax design shapes investment incentives. The 2010–2013 HST interlude was not a technical accident—it was the culmination of decades of provincial fiscal policy and, ultimately, democratic reversal. We trace that history in Section 2.1 and Section 2.2, covering more than seven decades of sales tax evolution in BC. In Section 2.3 we develop the user cost of capital framework that provides the theoretical mechanism linking tax design to investment decisions. Section 2.4 then unpacks the PST's complex exemption structure, which we argue is a key source of the heterogeneous effective burdens documented later.

2.1 From PST to HST and Back

Provincial sales taxes in Canada traditionally take the form of retail sales taxes (RSTs) levied on final sales of goods and, in some cases, services. Unlike value-added taxes (VAT) or harmonised sales taxes (HST), RSTs are generally non-creditable: firms pay the tax on intermediate inputs, including machinery and equipment, and cannot recover it through the production chain. This creates a *tax cascade*, whereby capital costs are taxed multiple times as goods move through production stages. The cumulative effect raises effective tax rates on investment above the statutory rate, discouraging capital formation.

The HST, which BC adopted in July 2010, is a unified value-added tax jointly administered with the federal Goods and Services Tax (GST). Because it allows full input credits, the HST eliminates the cascading effect on business inputs. A firm purchasing machinery under the HST pays the tax at the point of purchase but recovers it through credits against its own HST remittances, so the net cost of capital is unaffected by the tax.

Following a referendum in September 2011 in which 54.7% of voters rejected the HST, BC legislated a return to the PST. The transition was completed in April 2013. The PST reintroduction thus raised the user cost of capital for BC businesses by approximately 7 percentage points relative to the HST period (Lammam 2016, Dahlby 2024).

2.2 Historical Policy Chronology (1948–2026)

British Columbia's sales tax system has evolved considerably over more than seven decades, shaped by fiscal pressures, intergovernmental harmonisation efforts, and direct democratic intervention. Table 1

Table 1: British Columbia Sales Tax Policy Milestones, 1948–2026

Year	Policy	Key change and implication for M&E investment
1948	SST introduced	3% retail sales tax (SSTA); no input credits; M&E fully taxable.
1974	Rate increase	SST rate raised to 5%.
1987	Rate increase	SST rate raised to 6%.
1993	Rate increase	SST rate raised to 7%; remains at this level until 2010.
2010	HST adopted	July 1: PST replaced by 12% HST (GST+PST); full input tax credits eliminate effective tax on M&E.
2011	Referendum	September: 54.73% vote to repeal HST; Legislature commits to PST restoration.
2013	PST reintroduced	April 1: 7% PST reinstated under new PSTA; M&E again non-creditedly taxable. Raises user cost of capital by ≈ 7 percentage points relative to HST period.
2014	PSTA amendment	Rules for real property contracts clarified; minor exemption adjustments.
2019	Clean energy	Expanded PST exemptions for qualifying clean energy equipment.
2021	Digital products	PST extended to software, digital services, and related products.
2024	Software services	Further amendments to PST on software and telecommunication services.

Notes: SST = Social Services Tax (also called PST); SSTA = *Social Service Tax Act*; HST = Harmonized Sales Tax; PST = Provincial Sales Tax; PSTA = *Provincial Sales Tax Act*. **Sources:** BC Ministry of Finance; Dahlby (2024); authors' compilation.

traces the key milestones in this history, from the introduction of the Social Services Tax in 1948 through the incremental amendments of the current PST regime. Understanding this longer arc is important for contextualising the quasi-experimental variation exploited in this report.

1948–2009: The Social Services Tax era. In 1948, British Columbia introduced the *Social Service Tax* (SST) under the *Social Service Tax Act* (SSTA), initially as a 3% levy on retail sales to fund provincial social services. The rate was raised to 5% in 1974, to 6% in 1987, and to 7% in 1993, where it remained until the HST transition in 2010. In common parlance, the tax became known as the Provincial Sales Tax (PST). The PST applied to most tangible personal property; no input tax credit mechanism existed. Purchases of machinery and equipment by businesses were taxable and the tax paid could not be recovered, raising the effective cost of capital investment. Partial exemptions were introduced for certain manufacturing equipment over the years, though these remained narrow and usage-dependent.

2010–2012: The Harmonized Sales Tax interlude. On July 1, 2010, British Columbia adopted the Harmonized Sales Tax (HST) at a combined rate of 12%—a 5% federal component (GST) and a 7% provincial component—replacing the PST. The HST is a value-added tax with full input tax credits: firms

could recover HST paid on capital inputs through credits against their own HST remittances, setting the effective marginal cost of M&E to zero under the HST. This represented a significant improvement in BC's investment climate relative to the PST regime (Lammam 2016, Dahlby 2024).

The HST transition was politically contentious. Critics argued the province had adopted the HST without a public mandate and that it shifted burdens onto consumers. A citizen-initiated recall petition triggered a binding mail-in referendum in September 2011, in which 54.73% of registered voters cast ballots to repeal the HST. The legislature subsequently enacted legislation to restore the PST, a process completed on April 1, 2013.

2013–present: PST reintroduction and incremental reform. The PST was reinstated on April 1, 2013, at the pre-2010 rate of 7% under the new *Provincial Sales Tax Act* (PSTA), which replaced the old Social Service Tax Act with updated and consolidated legislation. While the PSTA modernised the exemption structure somewhat—introducing cleaner definitions of taxable and exempt goods and services—the fundamental design remained unchanged: the PST is a retail sales tax without input tax credits. M&E purchased by businesses for use in production remains taxable unless specifically exempted, re-imposing the capital cost increase that the HST had eliminated.

Subsequent years have brought incremental amendments: clarification of real property contract rules (2014), expanded exemptions for qualifying clean energy equipment (2019), extension of the PST to software and digital products (2021), and further adjustments to software and telecommunication services exemptions (2024). Despite these adjustments, the non-creditable design of the PST has been preserved throughout.

The stark contrast between the HST period (2010–2012) and the PST regime (pre-2010 and post-2013) is central to this report's identification strategy. The policy change was sudden, province-wide, and clearly defined in time, allowing comparison of BC's investment trajectory to that of control provinces in a difference-in-differences framework.

2.3 The User Cost of Capital and Marginal Effective Tax Rates

The user cost of capital, c , is the rental price firms face when acquiring an additional unit of capital. Following Hall & Jorgenson (1967), it can be written as:

$$c = \frac{(r + \delta)(1 - \tau z)}{1 - \tau} \cdot (1 + t_s), \quad (1)$$

where r is the real interest rate, δ is the depreciation rate, τ is the corporate income tax rate, z is the present value of depreciation allowances, and t_s is the sales tax on capital purchases. Under the HST, $t_s = 0$ (firms recover the full amount); under the PST, $t_s \approx 0.07$, directly raising c . Sectors with higher investment volumes, faster capital turnover, or tighter margins are mechanically more exposed to this increase.

The Marginal Effective Tax Rate (METR) on investment summarises the combined tax burden imposed

by all relevant taxes on a marginal investment project. Evidence from [Dahlby \(2024\)](#) and [BC Commission on Tax Competitiveness \(2016\)](#) indicates that the PST accounts for a substantial share of BC's METR on capital—estimates suggest roughly two-thirds of BC's excess METR relative to comparable provinces derives from the non-creditable tax on M&E.

2.4 The PST's Complex Exemption Structure

The BC PST applies to most purchases of tangible personal property unless specifically exempted. Exemptions for M&E are narrow and usage-dependent: certain machinery may be exempt if used for specific processes within a plant, but taxable if used in other contexts, requiring firms to self-assess and remit based on their particular operations ([Dahlby 2024](#)). Contractors installing goods into real property may face additional complications; mixed contracts must be carefully disaggregated to identify taxable components; and vendors that both use products internally and sell them externally must distinguish the purpose of each purchase.

This complexity imposes compliance costs beyond the statutory rate and creates heterogeneous effective burdens across firms and industries. It also means that the economic incidence of the PST is not uniform: sectors with broader exposure to taxable M&E, or with less capacity to track and manage compliance requirements, face higher effective tax burdens. These institutional features motivate the sector-level analysis developed in this report.

3 Data and Investment Trends

Before turning to the formal estimates, we document the raw patterns in the data and assess whether they are broadly consistent with the hypothesis that the PST operates through capital costs. We draw on a province–industry–year panel from Statistics Canada covering 2001–2024, described in [Section 3.1](#). In [Section 3.2](#), we show that BC's per-worker investment underwent a visible moderation after 2013—a divergence that stands out relative to the aggregate cross-provincial trend. [Section 3.3](#) then reveals that this moderation was far from uniform: capital-intensive sectors such as Mining and Utilities show the sharpest divergence, while Services remained broadly unaffected. These descriptive patterns foreshadow and motivate the heterogeneity analysis in [Section 4](#).

3.1 Data Sources and Construction

The empirical analysis uses a province–industry–year panel covering the ten Canadian provinces over 2001–2024. Investment and capital stock data are from Statistics Canada's *Flows and Stocks of Fixed Non-Residential Capital by Industry and Asset* tables. Employment data are from the *Annual Employment by Industry* tables; real value added is from *Gross Domestic Product at Basic Prices by Industry*. All monetary values are expressed in constant 2017 Canadian dollars.

Industries are harmonised using two-digit NAICS codes for the broad sectoral analysis and three-digit NAICS codes for manufacturing subsectors. [Table 2](#) defines the main variables. Per-worker measures

are emphasised throughout to control for differences in provincial scale and labour force composition; capital deepening (investment per worker) is the primary outcome.

Table 2: Definition of Main Variables

Variable	Definition
M&E Investment	Gross fixed capital formation in machinery and equipment (constant 2017 CAD).
M&E per Worker	M&E investment divided by total employment in industry i , province p , year t .
Capital Stock per Worker	Net capital stock divided by employment.
Real GDP	Real value added at basic prices (constant 2017 CAD).
M&E Intensity	Ratio of M&E investment to real GDP.
Capital Growth	Year-on-year growth rate of gross capital stock.

The sector classification used throughout the report is presented in Table B.1 in Appendix B.

3.2 Investment Trends Across Canadian Provinces

Figure 1 plots total M&E investment across provinces. BC, shown in black, follows a steadily rising aggregate trajectory with limited cyclical volatility. Alberta displays pronounced boom–bust dynamics driven by commodity cycles; Ontario records the highest absolute levels; Quebec maintains higher levels than BC throughout the sample.

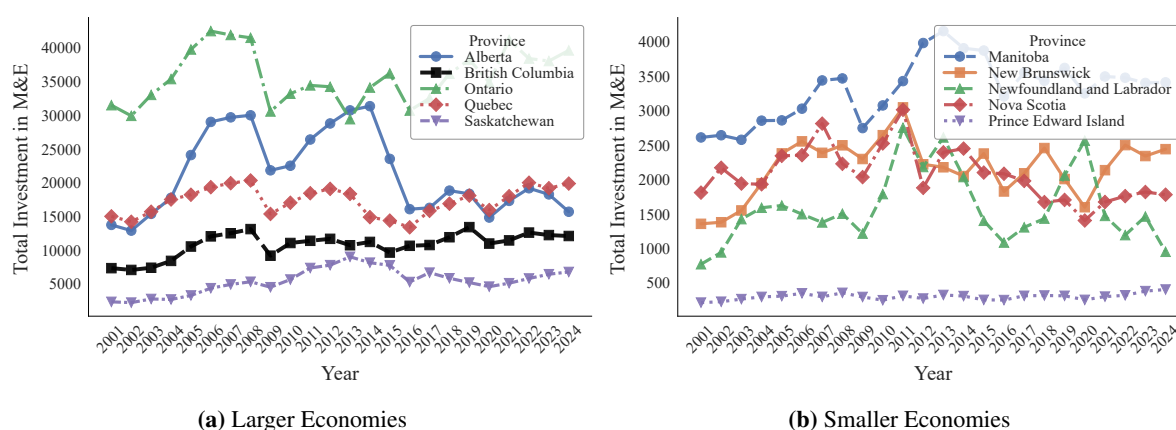


Figure 1: Total M&E Investment Across Canadian Provinces, 2001–2024

Notes: Total investment in machinery and equipment (millions of 2017 constant Canadian dollars). Panel (a): Alberta, BC, Ontario, Quebec, Saskatchewan. Panel (b): Manitoba, New Brunswick, Newfoundland and Labrador, Nova Scotia, Prince Edward Island. **Sources:** Statistics Canada; authors' calculations.

Normalising by employment reveals a different picture. Figure 2 shows M&E investment per worker. BC consistently invests more per worker than Quebec but trails Alberta. Notably, per-worker investment in BC does not recover its pre-2008 peak, exhibits a moderate rebound during 2009–2011 coinciding with the HST transition, and then declines between 2013 and 2015 following the PST reinstatement. Thereafter, capital deepening remains subdued. This timing is consistent with the PST operating through

the user cost channel.

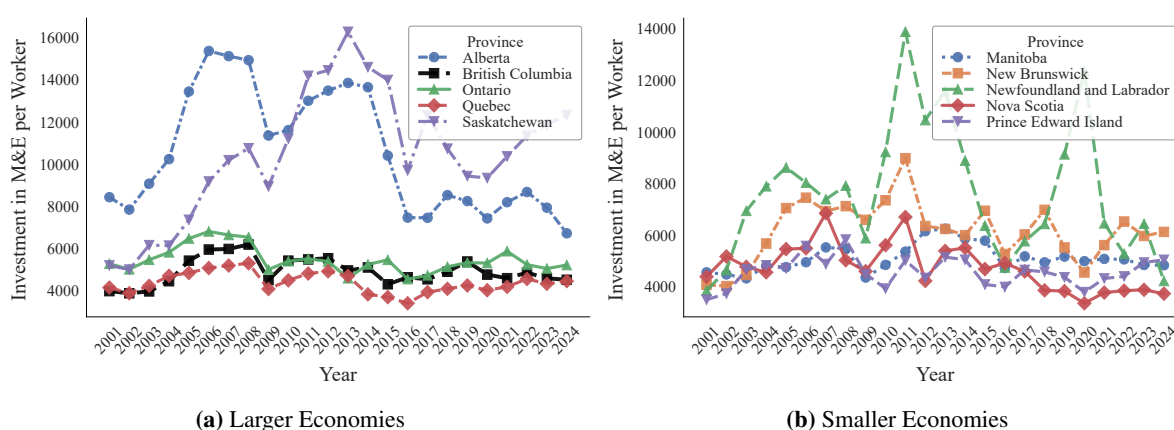


Figure 2: M&E Investment per Worker Across Canadian Provinces, 2001–2024

Notes: M&E investment per worker (2017 constant CAD). Panel (a): larger economies; Panel (b): smaller economies. **Sources:** Statistics Canada; authors' calculations.

Expressing M&E investment as a share of GDP (Figure 3) confirms a broadly declining trend across all provinces since the early 2010s, with BC allocating a smaller and steadily shrinking share of its output to capital formation since 2008.

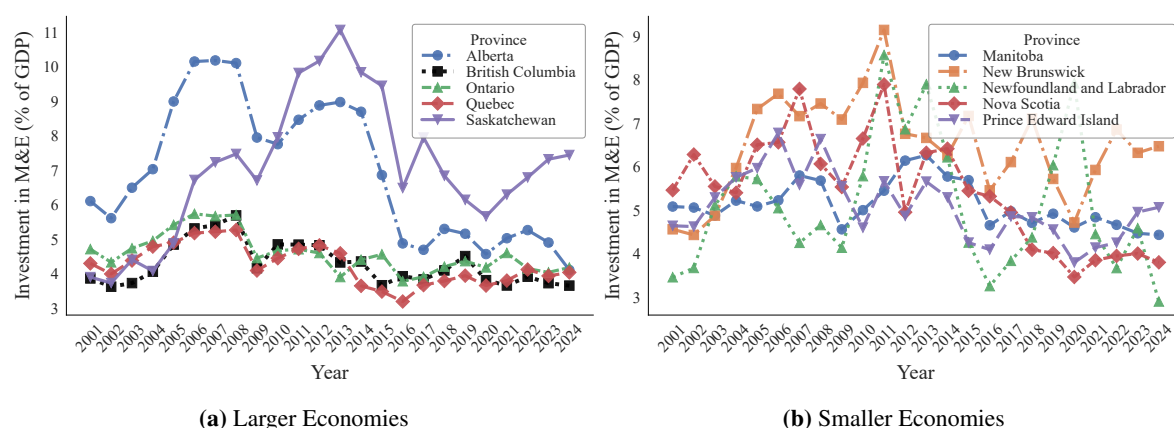


Figure 3: Share of M&E Investment in Provincial GDP, 2001–2024

Notes: Ratio of M&E investment to real GDP for two groups of provinces. **Sources:** Statistics Canada; authors' calculations.

3.3 Sectoral Heterogeneity in Capital Deepening

Because the PST operates through the cost of capital inputs, its effects should be strongest in capital-intensive sectors. Figure 4 compares M&E investment per worker in goods-producing and services sectors across provinces. Goods-producing industries consistently invest more per worker than services in every province and year, reflecting structural differences in production technology. Cyclical volatility is concentrated in goods sectors, especially in resource-intensive provinces.

Figure 5 disaggregates the goods block into Agriculture, Manufacturing, Construction, Utilities, Mining, and contrasts them with the aggregate Services sector. Mining and Utilities are clearly the most capital-

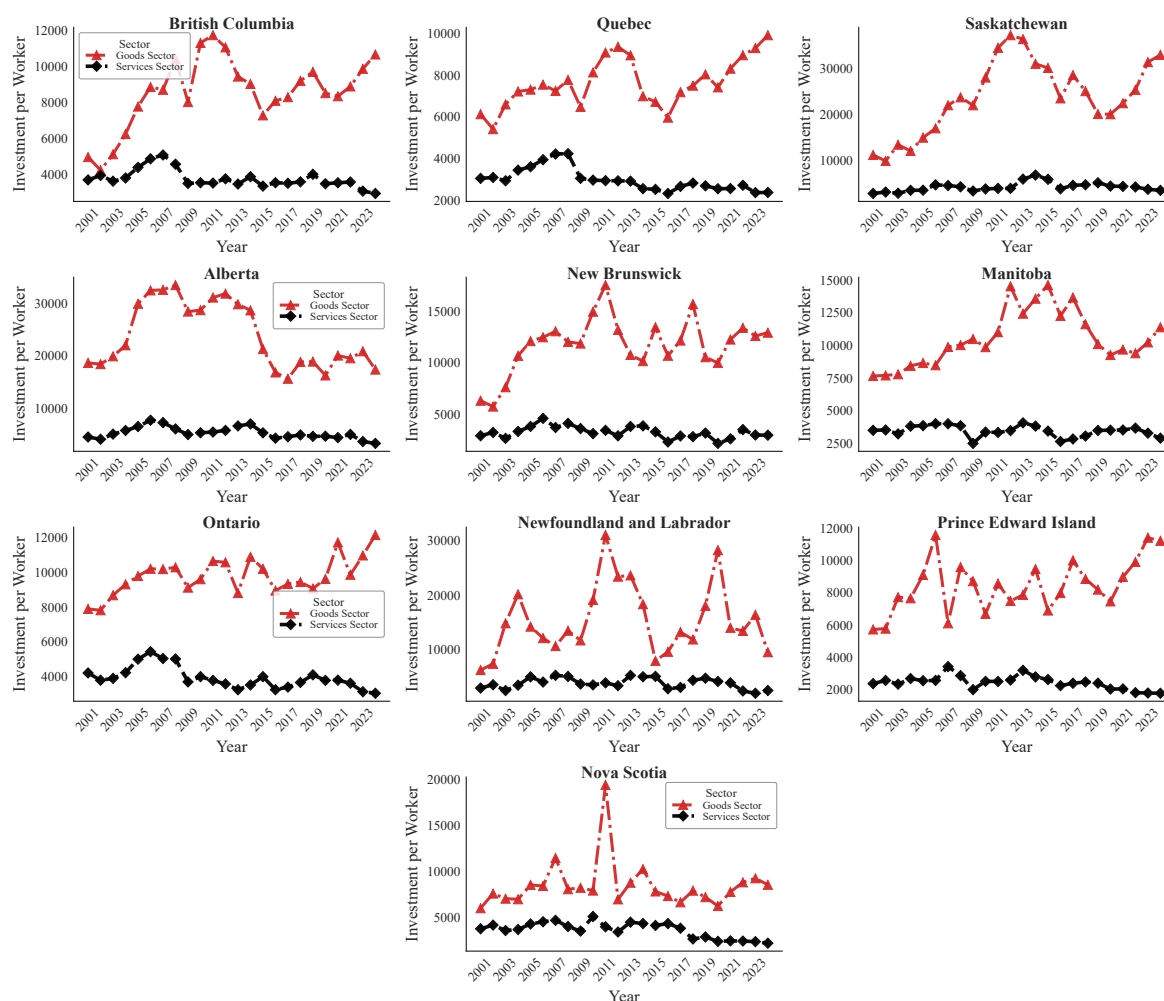


Figure 4: M&E Investment per Worker: Goods vs. Services Sectors, 2001–2024

Notes: Investment per worker (2017 constant CAD) in goods-producing (red) and services (black) sectors. Sector definitions follow Table B.1. **Sources:** Statistics Canada; authors' calculations.

intensive sectors. Within BC, mining and utilities experience marked contractions after 2012–2013, which contributes to the moderation visible in the aggregate goods block.

Table 3 compares average per-worker investment before (2001–2012) and after (2013–2024) the PST reintroduction in BC and in the Rest of Canada. Aggregate declines are comparable (–6.0% in BC vs. –5.0% in the rest), but the sectoral composition differs sharply: Mining contracts far more in BC (–39.3%) than elsewhere (–15.8%), and Utilities falls more in BC (–20.8%) than in other provinces (–2.7%). These pre/post comparisons are suggestive, but—as discussed in the econometric section—proper identification requires controlling for common macroeconomic shocks and persistent provincial differences.

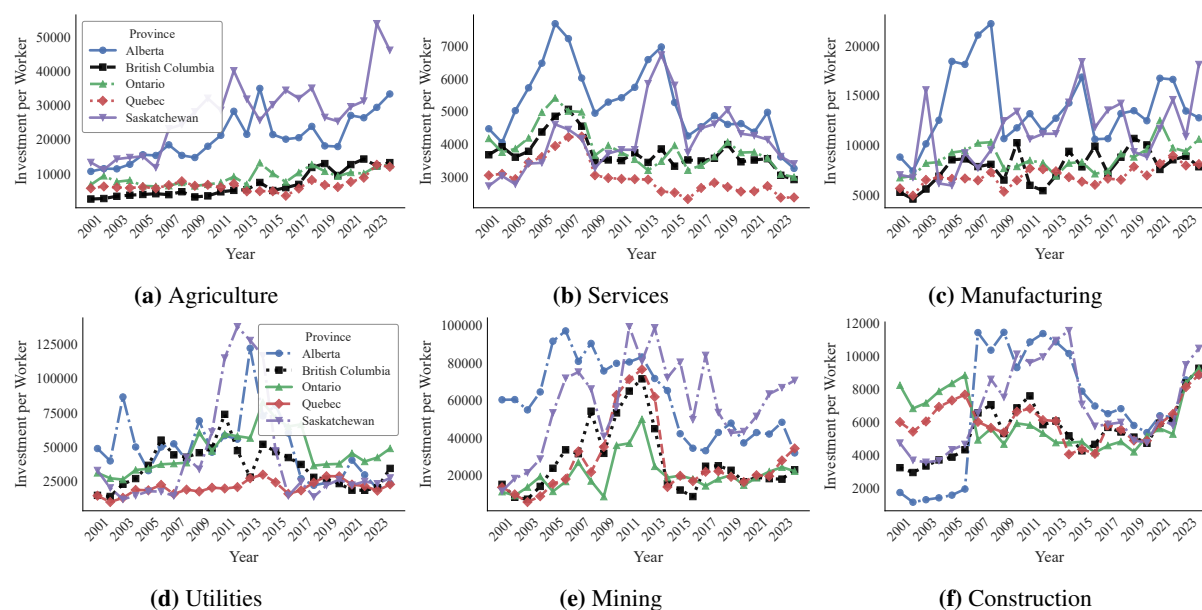


Figure 5: M&E Investment per Worker by Broad Sector, Larger Provinces, 2001–2024

Notes: Investment per worker in 2017 constant CAD. BC shown in black; other provinces in lighter shades. See Table B.1 for sector definitions. **Sources:** Statistics Canada; authors' calculations.

Table 3: Per-Worker M&E Investment by Sector: Rest of Canada vs. British Columbia

Sector	Period	Rest of Canada			British Columbia		
		<i>N</i>	Mean (SD)	$\Delta\%$	<i>N</i>	Mean (SD)	$\Delta\%$
Aggregate	2001–2012	108	6,726 (2,926)		12	5,052 (856)	
	2013–2024	108	6,387 (2,831)	–5.0	12	4,749 (296)	–6.0
Agriculture	2001–2012	108	9,422 (6,901)		12	3,829 (784)	
	2013–2024	108	14,198 (9,957)	+50.7	12	9,809 (3,465)	+156.2
Construction	2001–2012	138	6,120 (2,908)		12	5,064 (1,671)	
	2013–2024	139	6,210 (2,234)	+1.5	12	5,897 (1,488)	+16.4
Manufacturing	2001–2012	108	10,368 (7,373)		12	6,997 (1,701)	
	2013–2024	108	11,039 (5,163)	+6.5	12	8,655 (1,200)	+23.7
Mining	2001–2012	104	38,828 (47,254)		12	34,123 (22,103)	
	2013–2024	115	32,690 (34,018)	–15.8	12	20,721 (9,084)	–39.3
Services	2001–2012	144	4,735 (2,319)		12	4,007 (558)	
	2013–2024	144	3,992 (1,714)	–15.7	12	3,477 (291)	–13.2
Utilities	2001–2012	107	35,628 (29,441)		12	39,332 (17,551)	
	2013–2024	110	34,651 (25,236)	–2.7	12	31,157 (11,279)	–20.8

Notes: Mean per-worker investment (2017 constant CAD); standard deviations in parentheses. $\Delta\%$ is the percentage change in means from the pre-period (2001–2012) to the post-period (2013–2024). *N* is the total number of observations. **Sources:** Statistics Canada; authors' calculations.

Key Finding

The raw data reveal sharp sectoral disparities in BC's investment performance before and after 2013. Mining contracted by 39% per worker in BC versus 16% in the rest of Canada; Utilities fell by 21% in BC versus 3% elsewhere. These patterns are consistent with greater exposure to capital taxation in high-intensity sectors, but formal identification requires econometric controls for common shocks and provincial heterogeneity.

These descriptive patterns are suggestive, but a simple before-after comparison cannot identify the causal effect of the PST. At least two forces can produce divergent investment paths across provinces and sectors independently of any tax change. First, common macroeconomic shocks—commodity price cycles, the aftermath of the 2008–2009 global recession, and the sharp oil price correction of 2014–2016—hit resource-intensive provinces unevenly. An observed contraction in BC's mining or utilities investment could partly reflect these aggregate forces rather than the PST alone. Second, and more fundamentally, ongoing structural change in the Canadian economy continuously reshapes the sectoral composition of provincial labour forces, altering per-worker investment ratios even absent any policy perturbation. As employment concentrates in services and as capital-intensive industries pursue capital-for-labour substitution, sectoral differences in equipment investment per worker will widen over time through purely structural channels (Kuznets 1967, Baumol 1967, Herrendorf et al. 2014, Avoumatsodo & Leunga Noukwe 2024).

We address these confounds directly in the econometric specification. The key structural change control is *lagged sectoral employment share*—the share of a province's total employment accounted for by sector i in the prior year. This variable absorbs the slow-moving compositional shifts in industrial structure that alter per-worker investment ratios independently of capital costs: a sector shrinking as a share of provincial employment will mechanically exhibit rising investment per worker even if aggregate investment is flat, and vice versa. Additional controls address complementary channels. *Lagged M&E capital stock* and *lagged capital growth* capture investment inertia—the well-documented persistence in capital accumulation paths (Bond & Van Reenen 2007). *Lagged real GDP per worker* controls for demand-side fluctuations that shift investment independently of the tax environment. Province, industry, and year fixed effects further absorb time-invariant provincial differences, sector-level investment norms, and aggregate macroeconomic trends. Technical details of the econometric design are provided in Appendix A.

4 Econometric Evidence on the Investment Effects of the PST

The identification strategy is a difference-in-differences (DiD) design¹. We compare the evolution of M&E investment per worker in BC before and after April 2013 to the contemporaneous evolution in the nine other Canadian provinces—a control group that experienced no comparable change in retail sales tax treatment of capital inputs over this period. The difference of these two differences recovers the causal effect of the PST reintroduction under the assumption that, absent the reform, BC's investment

¹See Appendix A for specification and more details.

trajectory would have followed a trend parallel to that of the control group. We argue this assumption is credible: the reform was sudden, province-wide, and driven by a democratic vote rather than by investment conditions, making reverse causality implausible. The robustness of the PST coefficient to progressively richer control sets, documented below, provides supporting evidence.

We find that the PST reintroduction had a negative and statistically significant average effect of approximately 6 per cent on M&E investment per worker. Yet this aggregate figure conceals a great deal. Capital-intensive sectors bore the brunt of the adjustment: Mining lost 27.9 per cent and Utilities 25.2 per cent, while Services were essentially unaffected. Within manufacturing, traditional subindustries—those most reliant on taxable equipment and with the thinnest margins—fared worse than their advanced counterparts. We document these patterns in turn.

4.1 Average Effect of the PST Reintroduction

Table 4 presents the baseline difference-in-differences estimates from four progressively richer specifications. In Column (1), with controls for lagged M&E stock and lagged capital growth, the PST coefficient is -0.063 and statistically significant at the 5% level, implying an approximate 6 per cent reduction in M&E investment per worker in BC following the 2013 PST reintroduction.

Column (2) adds lagged GDP per worker; the estimated effect is -0.062 , significant at 10%. Column (3) additionally controls for an interaction between Alberta and oil prices to absorb province-specific macroeconomic shocks in resource sectors; the PST coefficient is -0.066 and significant at 5%. Column (4), the preferred specification, further controls for lagged employment share; the estimated effect is -0.063 , significant at 5%.

The stability of the coefficient across all four specifications—and the consistently negative and statistically significant sign—supports the interpretation that the PST reintroduction causally reduced M&E investment per worker in BC.

Key Finding

Across all specifications and control sets, the 2013 PST reintroduction is associated with a statistically significant **6 per cent reduction** in M&E investment per worker in BC relative to other Canadian provinces.

4.2 Differential Effects Across Sectors

The average 6% reduction masks substantial variation across industries. Table 5 presents sector-specific estimates. Column (4) is the preferred specification and includes lagged M&E growth, M&E intensity, capital per worker, and lagged employment share as controls.

Table 6 computes the total PST effect for each sector by adding the Agriculture baseline ($\hat{\beta}_{agr} = 0.304$) to the sector-specific interaction coefficient². These totals represent the net change in log M&E investment per worker in BC, post-2013, relative to the counterfactual (what would have happened absent the PST).

²see Appendix A for the derivation of marginal effect. Approximate standard errors for the total effect are computed as

Table 4: Baseline Effect of BC PST on M&E Investment per Worker

Variable	<i>Dependent variable: log(M&E investment per worker)</i>			
	(1)	(2)	(3)	(4)
PST (treatment indicator)	−0.063** (0.024)	−0.062* (0.022)	−0.066** (0.026)	−0.063** (0.024)
Lagged M&E stock	0.566*** (0.040)	0.599*** (0.041)	0.603*** (0.040)	0.630*** (0.040)
Lagged capital growth	2.555*** (0.211)	2.535*** (0.222)	2.524*** (0.217)	2.454*** (0.222)
Lagged GDP per worker		0.376*** (0.099)	0.380*** (0.098)	0.301** (0.114)
Alberta × Oil price			−0.0018** (0.001)	−0.0019** (0.001)
Lagged employment share				−6.845** (1.603)
Province FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Observations	3,388	3,383	3,383	3,383
R^2	0.938	0.940	0.940	0.941
Within R^2	0.303	0.321	0.321	0.327

Notes: Estimates of the baseline DiD specification. The PST indicator equals 1 for BC from 2013 onward and 0 otherwise. Observations for all provinces in 2010–2012 (HST transition period) are excluded from the estimation sample. Column (4) is the preferred specification. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6 presents a ranked summary of total effects, making the ordering of sectoral impacts immediately visible.

The findings reveal a clear gradient tied to capital intensity:

- **Mining** (−27.9%) and **Utilities** (−25.2%) bear the heaviest burden, consistent with their intensive reliance on taxable capital equipment. These sectors face both large absolute capital costs and narrower exemptions under the PST structure (Dahlby 2024).
- **Construction** records a moderate, marginally significant decline (−6.1%), partly offset by strong demand in BC's housing and infrastructure sectors.
- **Services** show no significant net change (+1.2%, not significant), consistent with the sector's comparative independence from M&E-intensive production.

$\left[SE(\hat{\beta}_{agr})^2 + SE(\hat{\theta}_k)^2\right]^{0.5}$. This treats the two estimates as independent. % Change $\approx \left[\exp(\hat{\beta}_{agr} + \hat{\theta}_k) - 1\right] \times 100$.

Table 5: PST Effects on M&E Investment per Worker by Broad Sector

Variable	<i>Dependent variable: log(M&E investment per worker)</i>			
	(1)	(2)	(3)	(4)
PST × Agriculture (ref.)	0.473*** (0.035)	0.430*** (0.018)	0.473*** (0.029)	0.304*** (0.044)
PST × Construction	−0.457*** (0.017)	−0.633*** (0.014)	−0.580*** (0.017)	−0.367*** (0.041)
PST × Manufacturing	−0.641*** (0.006)	−0.365*** (0.023)	−0.363*** (0.019)	−0.129*** (0.057)
PST × Mining	−0.747*** (0.026)	−0.719*** (0.022)	−0.770*** (0.022)	−0.631*** (0.063)
PST × Services	−0.501*** (0.030)	−0.456*** (0.026)	−0.473*** (0.024)	−0.292*** (0.039)
PST × Utilities	−0.897*** (0.017)	−0.820*** (0.017)	−0.757*** (0.012)	−0.594*** (0.028)
Lagged M&E growth	3.112***	2.575***	2.419***	2.325***
Lagged M&E intensity		2.681***	2.095***	2.043***
Lagged capital per worker			0.632***	0.695***
Lagged employment share				−0.385***
Lagged employment share × Sector				✓
Province FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Observations	3,383	3,383	3,383	3,383
R^2	0.927	0.936	0.943	0.944
Within R^2	0.178	0.274	0.357	0.368

Notes: All interaction coefficients are reported relative to Agriculture (reference category). The total PST effect for sector k equals the Agriculture coefficient plus the interaction term for sector k ; see Table 6 for computed totals. Column (4) is the preferred specification. Observations for all provinces in 2010–2012 are excluded. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

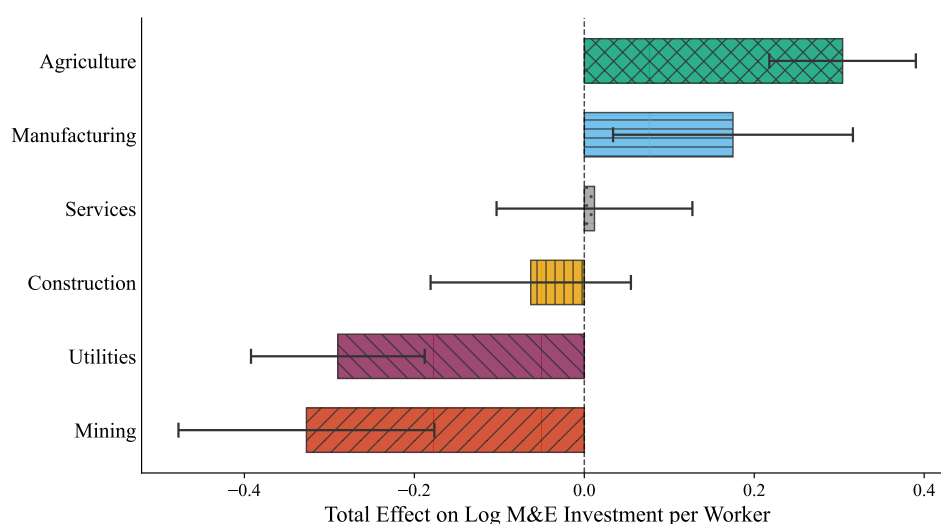
- **Agriculture** records a large positive effect (+35.5%), which reflects a strong divergence in BC's agricultural investment trajectory relative to other provinces over this period; this may capture structural changes specific to BC's agricultural sector and serves as the reference category in the estimation.
- **Manufacturing** shows a positive total effect (+19.1%, significant), meaning that BC's manufacturing investment per worker increased *more* than in other provinces after 2013 on average. This result is consistent with the descriptive evidence (BC manufacturing investment grew 23.7% post-2013 vs. 6.5% in the rest of Canada). It may reflect sector-specific exemptions, demand-side shifts, or com-

Table 6: Total PST Effects by Sector: Interaction Coefficient and Total Effect

Sector	Interaction term		Total effect ($\hat{\beta}_{agr} + \hat{\theta}_k$)		% Change
	$\hat{\theta}_k$	(SE)	Total	(approx. SE)	
Agriculture	—	—	+0.304***	(0.044)	+35.5%
Services	−0.292***	(0.039)	+0.012	(0.059)	+1.2%
Construction	−0.367***	(0.041)	−0.063	(0.060)	−6.1%
Utilities	−0.594***	(0.028)	−0.290***	(0.052)	−25.2%
Mining	−0.631***	(0.063)	−0.327***	(0.077)	−27.9%
Manufacturing	−0.129***	(0.057)	+0.175**	(0.072)	+19.1%

Notes: This table shows the total PST effect by sector. Interaction coefficients $\hat{\theta}_k$ are from Column (4) of Table 5. The total effect equals $\hat{\beta}_{agr} + \hat{\theta}_k$ where $\hat{\beta}_{agr} = 0.304$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Sources: Authors' calculations.

**Figure 6:** Total PST Effects on M&E Investment per Worker by Sector

Notes: Bars show log-point regression coefficients—the estimated effect of the PST on log M&E investment per worker by sector (Column 4 of Table 5). Sectors are ranked from most adverse to most beneficial total effect. Error bars show 95% confidence intervals. Implied percentage changes are computed as $(\exp(\hat{\beta}_{agr} + \hat{\theta}_k) - 1) \times 100$. Sources: Authors' calculations.

positional changes within manufacturing. However, within manufacturing, traditional subindustries are more adversely affected than advanced ones, as discussed in the next section.

4.3 Heterogeneity Within Manufacturing

Table 7 and Figure 7 examine whether advanced and traditional manufacturing subindustries respond differently to the PST. Advanced manufacturing (machinery, electrical equipment, chemicals, transportation equipment, plastics and metals) serves as the reference group; traditional manufacturing (food and beverage, textiles, wood products, paper, furniture, printing) is interacted with the PST indicator.

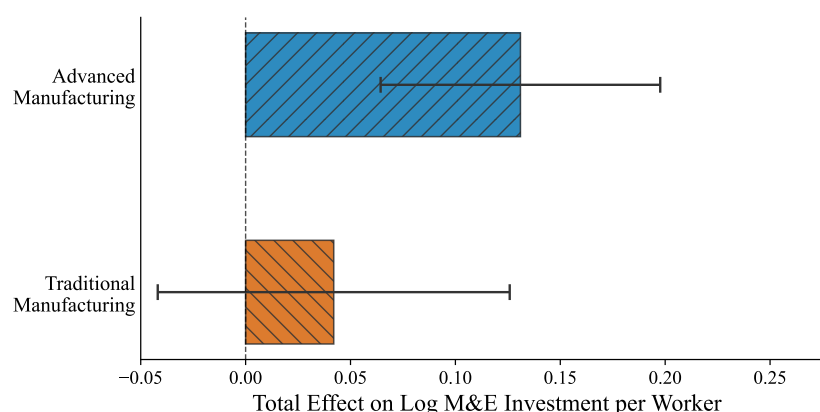
The interaction term for traditional manufacturing is negative and statistically significant across all three columns. The Column (4) estimate implies that traditional manufacturing experienced an additional 8.9

Table 7: PST Effects Within Manufacturing: Advanced vs. Traditional Subindustries

Variable	<i>Dependent variable: log(M&E investment per worker)</i>			
	(1)	(2)	(3)	(4)
PST (Advanced Manufacturing, ref.)	0.165*** (0.045)	0.148** (0.044)	0.147*** (0.043)	0.131** (0.044)
PST × Traditional Manufacturing	−0.094** (0.041)	−0.075*** (0.030)	−0.099*** (0.029)	−0.089** (0.030)
Lagged capital per worker	0.611***	0.679***	0.617***	0.571***
Lagged M&E intensity		1.575***	1.409***	1.441***
Lagged sector GDP			0.105*	0.226**
Lagged employment share				−0.170*
Lagged employment share × Sector				✓
Province FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Observations	2,495	2,495	2,495	2,495
R^2	0.654	0.677	0.681	0.683
Within R^2	0.188	0.242	0.252	0.256

Notes: Estimation of the within-manufacturing specification. Advanced manufacturing is the reference group; the total effect for traditional manufacturing equals $0.131 + (-0.089) = +0.042$ (Column 4), not significantly different from zero. Observations for all provinces in 2010–2012 are excluded. Column (4) is preferred. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

log-point reduction in investment per worker relative to advanced manufacturing. Total effects (preferred specification) are: Advanced, +13.1%; Traditional, +4.2% (not significantly different from zero).

**Figure 7:** PST Effects Within Manufacturing: Advanced vs. Traditional Subindustries

Notes: Bars show log-point regression coefficients—the estimated total effect of the PST on log M&E investment per worker (Column 4 of Table 7). Error bars show 95% confidence intervals.

Sources: Authors' calculations.

Key Finding

Traditional manufacturing subindustries (food and beverage, wood products, paper, textiles) are significantly more adversely affected by the PST than advanced manufacturing (machinery, electrical equipment, chemicals). This is consistent with tighter margins, greater capital cost sensitivity, and lower technological flexibility in traditional sectors.

5 Policy Implications

Evidence alone does not reform policy—it must be translated into recommendations that policymakers can act on. We argue that the PST's non-creditable design imposes a genuine and avoidable cost on BC's investment climate, with consequences that compound over time through slower capital deepening and weaker productivity growth. Five concrete reform recommendations follow in Section 5.2, ranging from targeted input tax credits to a broader re-evaluation of BC's sales tax architecture. We conclude in Section 5.3 by placing these findings in the wider context of interprovincial competitiveness—a dimension that grows more consequential as capital becomes increasingly mobile.

5.1 The Investment Cost of the PST

The evidence establishes that BC's non-creditable PST imposes a measurable and economically significant cost on business investment in machinery and equipment. The mechanism is straightforward: by raising the effective price of capital inputs, the tax reduces firms' incentives to invest, particularly in sectors that rely heavily on such inputs. Capital deepening—investment per worker in productive assets—is a principal driver of labour productivity growth in both neoclassical and endogenous growth models (Solow 1956). A tax that persistently suppresses this margin therefore carries long-run costs in productivity, wages, and competitiveness that go beyond the direct revenue it generates.

The PST's complex exemption structure amplifies these costs. Because exemptions are usage-dependent and contractual-context-specific, firms must devote resources to compliance tracking, self-assessment, and audit risk management. These administrative costs are asymmetric: larger firms can spread them across a broader tax base, while smaller firms in traditional manufacturing or mining bear a proportionally higher burden.

5.2 Five Recommendations for Policymakers

Recommendation 1: Expand M&E Input Tax Credits

Introduce a full or partial credit against PST paid on machinery and equipment purchases for registered businesses. This would directly reduce the effective cost of capital without eliminating the PST for consumer goods, preserving revenue from final consumption while removing the most distortionary aspect of the current regime.

Recommendation 2: Simplify the Exemption Structure

Consolidate and clarify PST exemptions for capital inputs. Clear, objective criteria for which M&E purchases qualify for exemption—rather than usage-based or contract-type-based rules—would reduce compliance costs, decrease the risk of errors and penalties, and create a more level playing field across firm sizes. Engaging directly with industry associations in Mining, Utilities, and traditional Manufacturing would help identify the most burdensome compliance requirements.

Recommendation 3: Prioritise Capital-Intensive Sectors

The evidence shows that Mining and Utilities bear the largest investment distortions (−27.9% and −25.2% respectively). Any targeted PST relief for capital goods should be designed with these sectors in mind, as the return on reducing effective tax rates is highest where the distortion is largest.

Recommendation 4: Evaluate a Return to a Value-Added Tax Framework

The most direct route to eliminating the cascading distortions of the PST would be to re-harmonise BC's sales tax with the federal GST. Under an HST, businesses would recover all sales taxes paid on inputs, setting the effective tax burden on capital to zero. BC's experience during 2010–2012 provides evidence of what such a shift could achieve. While the political economy of re-harmonisation is complex, the economic case is clear: Ontario, New Brunswick, Nova Scotia, Newfoundland, and Prince Edward Island already operate under harmonised regimes and benefit from lower effective METRs on investment.

Recommendation 5: Build Industry-Level Tax Burden Panel Data

Current evidence is based on aggregate investment data and relies on indirect inference about effective tax rates. Detailed sector-level data on the actual PST paid by firms, obtainable through collaboration with the BC Ministry of Finance, would allow policymakers to measure effective tax burdens directly, evaluate the incidence of incremental tax changes, and design reform packages targeted at the most distorted sectors. Such data would also strengthen future research on the long-run effects of capital taxation on productivity and wages.

5.3 The Competitiveness Dimension

BC does not make investment decisions in isolation. Firms operate in a national and international capital market. Evidence from [Clemens et al. \(2008\)](#) and [Mintz & Bazel \(2021\)](#) shows that Canada's effective tax rate on new investment already exceeds that of the United States. Within Canada, BC's PST places it at a disadvantage relative to HST provinces. [Lammam \(2016\)](#) calculate that the PST materially raises BC's METR relative to the HST period and to comparable provinces. This tax disadvantage may affect both extensive-margin decisions (where to locate a facility) and intensive-margin decisions (how much to invest at an existing location). The econometric evidence in this report speaks primarily to the intensive margin, but the aggregate effects on location choice and industrial structure may be larger still.

6 Conclusion

The evidence in this report establishes that British Columbia's 2013 reversion to the Provincial Sales Tax reduced M&E investment per worker by approximately 6 per cent—a causal estimate that holds across four progressively richer specifications, survives the inclusion of controls for investment inertia, demand conditions, and structural change, and is robust to the choice of estimation sample. The result is precisely what the user cost of capital model predicts ([Hall & Jorgenson 1967](#)): a non-creditable retail sales tax raises the effective price of capital inputs, and firms respond by investing less. The magnitude implies that roughly one in seventeen dollars of M&E investment per worker was forgone relative to the counterfactual of continued HST coverage.

The sectoral pattern deepens this picture in a theoretically coherent way. The PST operates through capital costs, so its distortionary burden should be heaviest where capital intensity is greatest and exemptions are narrowest. Mining and Utilities satisfy both criteria: they are the most equipment-intensive sectors in BC's economy and receive limited relief under the PST's usage-based exemption structure. The estimated declines of 27.9 per cent and 25.2 per cent respectively reflect both large absolute cost increases and limited scope for substitution away from taxable M&E. Services, which invest relatively little per worker in taxable equipment, show no significant response—precisely as the theory predicts. The sectoral gradient is not noise; it is the signature of the user cost mechanism at work.

Within manufacturing, the distinction between advanced and traditional subindustries adds a layer of resolution that aggregate manufacturing figures obscure. Traditional manufacturing—food and beverage, wood products, paper, and textiles—was significantly more adversely affected by the PST than advanced manufacturing: the differential effect amounts to 8.9 log points and is statistically significant across all specifications. Sectors operating with older capital vintages, thinner margins, and lower technological flexibility bear a heavier burden from a capital cost increase, being less able to substitute toward untaxed inputs or to absorb higher costs through prices. This within-sector gradient reinforces the broader finding that the PST's distortionary impact is concentrated precisely where capital is most indispensable and firms are least insulated.

The policy implication is direct. BC's non-creditable PST architecture imposes a genuine and avoidable cost on business investment, concentrated in exactly the sectors—Mining, Utilities, and traditional Manufacturing—where capital deepening matters most for productivity and long-run growth. The five recommendations in [Section 5](#) offer a spectrum of responses: from targeted input tax credits that address the most distorted sectors within the current PST framework, to full harmonisation with the federal GST that would eliminate the cascading distortion at its source. The economic case for reform is clear. Whether BC chooses a targeted or systemic response, the evidence presented here provides the empirical foundation for that choice.

References

- Avoumatsodo, K. & Leunga Noukwe, I. (2024), 'Time-varying Input-Output Linkages and Structural Change'.
- URL:** <https://ssrn.com/abstract=5243498>
- Baumol, W. J. (1967), 'Macroeconomics of Unbalanced Growth: The Anatomy of Urban Crisis', *The American Economic Review* **57**(3), 415–426.
- BC Commission on Tax Competitiveness (2016), Improving British Columbia's business tax competitiveness, Technical report, Government of British Columbia.
- Bond, S. & Van Reenen, J. (2007), 'Microeconometric models of investment and employment', *Handbook of Econometrics* **6**, 4417–4498.
- Clemens, J. et al. (2008), *The Impact and Cost of Taxation in Canada: The Case for Flat Tax Reform*, Fraser Institute.
- Dahlby, B. (2024), The case for replacing British Columbia's inefficient PST with a made-in-BC VAT, Technical report, Fraser Institute.
- Hall, R. E. & Jorgenson, D. W. (1967), 'Tax policy and investment behavior', *American Economic Review* **57**(3), 391–414.
- URL:** <https://www.jstor.org/stable/1812110>
- Herrendorf, B., Rogerson, R. & Valentinyi, A. (2014), 'Growth and Structural Transformation', *Handbook of Economic Growth* **2**, 855–941.
- Kuznets, S. (1967), *Modern economic growth : rate, structure and spread.*, New Haven ; London : Yale University Press.
- Lammam, C. (2016), Submission to British Columbia's commission on tax competitiveness, Technical report, Fraser Institute.
- Mintz, J. M. & Bazel, P. (2021), 2020 tax competitiveness report: Canada's investment challenge, Technical Report 1, The School of Public Policy Publications.
- Solow, R. M. (1956), 'A contribution to the theory of economic growth', *Quarterly Journal of Economics* **70**(1), 65–94.

A Econometric Methodology

This appendix provides a self-contained technical account of the econometric strategy underlying the results in Section 4. The central challenge is to isolate the causal effect of the 2013 PST reintroduction on sectoral M&E investment from the many other forces driving investment dynamics across provinces and industries—common macroeconomic shocks, investment persistence, demand conditions, and ongoing structural change in provincial employment composition. We address this by exploiting the discrete, province-specific policy change in a difference-in-differences framework, augmented with progressively richer controls. The three specifications used in the main analysis are presented in turn, followed by a discussion of the identifying assumptions and the evidence in their support.

A.1 Difference-in-Differences Specification

The data are organised as a three-dimensional panel: industry i , province p , and year t . The panel covers all ten Canadian provinces, all two-digit NAICS sectors, and years 2001–2024, with the 2010–2012 HST transition period excluded for BC. The dependent variable is the natural log of M&E investment per worker. The log transformation compresses the right skew in investment distributions and allows the coefficient on the treatment indicator to be interpreted as an approximate percentage change.

Let i index industries, p provinces, and t years. The baseline specification is:

$$Y_{ipt} = \alpha + \beta \text{PST}_{pt} + \gamma \mathbf{X}_{ipt-1} + \delta_i + \mu_p + \lambda_t + \varepsilon_{ipt}, \quad (2)$$

where Y_{ipt} is the log of M&E investment per worker. The parameter of interest is β : under the parallel trends assumption, it recovers the average causal effect of the PST reintroduction on log investment per worker. The implied percentage change is $\exp(\hat{\beta}) - 1$.

Treatment indicator. The variable PST_{pt} equals 1 for British Columbia from 2013 onward and 0 for all other province-year cells. This binary indicator captures the April 2013 shift from the fully creditable HST back to the non-creditable PST. Because the reform was province-wide and affected all sectors simultaneously, the identifying variation is entirely within BC, contrasting its post-2013 investment trajectory to its own pre-treatment baseline (before 2010) relative to a common trend estimated from the control provinces.

Exclusion of the HST transition period. Observations for 2010–2012 are dropped for BC. Including these years would conflate two conceptually distinct changes: the investment response to the HST adoption in 2010, which eliminated the effective tax on M&E, and the later response to the PST reintroduction in 2013. Retaining the transition years in the pre-period would bias the pre-reform baseline upward, since investment rose during the HST interlude, thereby inflating the estimated PST effect. The baseline sample therefore covers 2001–2009 (pre-period) and 2013–2024 (post-period). Table 1 in Section 2 summarises the policy history.

Control variables. The lagged control vector $\mathbf{X}_{ipt-1}$ contains four groups of variables, each addressing

a distinct source of variation in investment beyond the PST:

- *Lagged M&E capital stock.* Investment is strongly mean-reverting around a stock target: firms with large accumulated stocks may invest less on the margin, while those with depleted stocks may invest more. Including the lagged stock controls for this capital adjustment motive and for heterogeneity in initial endowments across sector-province cells (Bond & Van Reenen 2007).
- *Lagged capital growth rate.* Year-to-year investment exhibits positive serial correlation beyond what the stock level explains—investment booms tend to persist before reverting. The lagged growth rate captures this momentum and reduces residual autocorrelation in the error term.
- *Lagged real GDP per worker.* Output conditions affect the demand for capital services independently of its cost. Provinces or sectors experiencing productivity gains or output expansions will invest more per worker regardless of the tax environment; including lagged GDP per worker prevents demand fluctuations from biasing the PST coefficient.
- *Alberta $\times$ log oil price.* Alberta's investment is dominated by oil and gas, which generates pronounced boom-bust cycles tied to global commodity prices. This interaction flexibly absorbs the Alberta-specific oil price sensitivity, so that the estimated year fixed effects are not driven by Alberta's resource cycle distorting the common trend.

In the preferred specification in column (4), the vector additionally includes *lagged sectoral employment share*—the fraction of total provincial employment in sector i in year $t - 1$. This variable absorbs slow-moving compositional shifts in provincial industrial structure that mechanically alter per-worker investment ratios: as employment concentrates in services relative to capital-intensive goods sectors, per-worker investment in those goods sectors rises even if aggregate investment is unchanged. Omitting this variable could attribute genuine structural-change dynamics to the PST treatment effect. All controls enter with a one-period lag to mitigate simultaneity bias.

Fixed effects. The three-way fixed-effects structure is central to identification:

- *Province fixed effects* (μ_p) absorb time-invariant differences in provincial investment climates—geography, natural resource endowments, and persistent tax differences unrelated to the PST reform.
- *Industry fixed effects* (δ_i) absorb permanent differences in capital intensity across sectors. Mining is structurally far more M&E-intensive than services regardless of the PST; failing to control for this would conflate structural heterogeneity with the treatment effect.
- *Year fixed effects* (λ_t) absorb macroeconomic shocks common to all provinces—recessions, global interest rate cycles, and secular trends in equipment investment. Together, the three sets of fixed effects ensure $\hat{\beta}$ is identified from within-province, within-industry, within-year variation.

Standard errors. Standard errors are clustered at the province level to allow for arbitrary serial correlation and heteroskedasticity within each province's investment series. This is the natural choice given

that treatment is assigned at the province level. With ten provinces in the sample, the number of clusters is small and cluster-robust standard errors can be imprecise in finite samples; readers should interpret marginal significance levels conservatively. The consistency of estimated effects across specifications with varying control structures provides additional reassurance beyond formal hypothesis tests.

A.2 Sectoral Heterogeneity Specification

The baseline specification (2) imposes a common PST effect across all sectors, which is restrictive given the heterogeneity in capital intensity documented in Section 3. We relax this by interacting the treatment indicator with sector dummies:

$$Y_{ipt} = \alpha + \beta_{\text{agr}} \text{PST}_{pt} + \sum_{k \neq \text{agr}} \theta_k (\text{PST}_{pt} \times \mathbb{1}_{i=k}) + \gamma \mathbf{X}_{ipt-1} + \delta_i + \mu_p + \lambda_t + \varepsilon_{ipt}. \quad (3)$$

Here β_{agr} measures the total PST effect in the reference sector, and each θ_k measures the *differential* effect in sector k relative to that baseline.

Choice of reference category. Agriculture is the reference sector. This is not arbitrary: under BC's PST, farm machinery and certain agricultural inputs benefit from specific input exemptions, making Agriculture the sector least expected to bear the full brunt of the non-creditable tax on M&E. Designating it as the reference produces $\hat{\theta}_k$ estimates that are negative for more exposed sectors and less negative (or positive) for more sheltered ones—a natural ordering tied to PST exposure.

Total effects and their standard errors. The total PST effect in sector k equals $\hat{\tau}_k = \hat{\beta}_{\text{agr}} + \hat{\theta}_k$. Its exact variance is:

$$\text{Var}(\hat{\tau}_k) = \text{Var}(\hat{\beta}_{\text{agr}}) + \text{Var}(\hat{\theta}_k) + 2 \text{Cov}(\hat{\beta}_{\text{agr}}, \hat{\theta}_k).$$

In fully interacted models, $\text{Cov}(\hat{\beta}_{\text{agr}}, \hat{\theta}_k)$ is typically negative: the interaction term absorbs part of the variance in the baseline coefficient, so the two estimates co-move negatively. Omitting the covariance therefore *overstates* the standard error. We report the conservative approximation $\widehat{\text{SE}}(\hat{\tau}_k) \approx \sqrt{\text{Var}(\hat{\beta}_{\text{agr}}) + \text{Var}(\hat{\theta}_k)}$, yielding confidence intervals wider than the exact ones. Any sector for which even these wider intervals exclude zero can be reported with confidence. Percentage changes are $(\exp(\hat{\tau}_k) - 1) \times 100$.

A.3 Manufacturing Heterogeneity

The broad-sector specification aggregates all manufacturing into a single category, potentially masking heterogeneity within the sector. We therefore estimate a separate within-manufacturing specification, restricting the sample to manufacturing subsectors and taking advanced manufacturing as the reference group:

$$Y_{ipt} = \alpha + \beta_A \text{PST}_{pt} + \theta_T (\text{PST}_{pt} \times \mathbb{1}_{i=T}) + \gamma \mathbf{X}_{ipt-1} + \delta_i + \mu_p + \lambda_t + \varepsilon_{ipt}. \quad (4)$$

Here β_A is the total PST effect in advanced manufacturing and θ_T is the additional differential effect in traditional manufacturing relative to advanced. The total effect for traditional manufacturing is $\hat{\beta}_A + \hat{\theta}_T$.

Rationale for the advanced/traditional split. The classification follows established conventions in industrial organisation and innovation research. Advanced manufacturing comprises machinery, electrical equipment, transportation equipment, chemicals, plastics, metals, petroleum and coal products, and non-metallic minerals—sectors with higher R&D intensity, greater capital-skill complementarity, and more sophisticated supply chains. Traditional manufacturing covers food and beverage, textiles, clothing, wood products, paper, printing, furniture, and miscellaneous manufacturing—sectors characterised by lower technological intensity, older capital vintages, and historically thinner margins. The user cost channel predicts greater distortion where capital is more complementary to production and where margins are too thin to absorb cost increases through price adjustment. The split is designed to test precisely this mechanism.

A.4 Identification Assumptions

Parallel trends. The causal interpretation of $\hat{\beta}$ rests on the *parallel trends* assumption: absent the 2013 PST reform, M&E investment per worker in BC would have followed a path parallel to that of the nine control provinces. Formally, this requires that the expected change in untreated potential outcomes is equal for BC and the control group in each post-reform period. This assumption cannot be directly tested since the counterfactual is unobserved, but several features of the design support its credibility.

First, *the reform was exogenous to investment conditions*. The 2013 reversion to the PST was determined by a 2011 democratic referendum driven by voter preferences over the distributional consequences of the HST for consumers—not by the investment outlook in BC. Reverse causality is therefore implausible.

Second, *the coefficient is stable across specifications*. Adding lagged GDP per worker, the Alberta oil price interaction, and the employment share control—each of which directly addresses a plausible confound—leaves $\hat{\beta}$ essentially unchanged across Columns (1) through (4). Under a parallel trends violation driven by these forces, the coefficient would shift when they are included; it does not.

Third, *the robustness tables are consistent*. Appendix C shows that including the 2010–2012 transition years yields qualitatively consistent estimates with somewhat smaller magnitude. This is precisely the pattern expected if the HST interlude represents a genuine positive investment shock that mildly lowers the estimated PST effect when the transition period is retained in the pre-period baseline.

Potential threats. The main threat to identification would be a BC-specific investment shock—unrelated to the PST—that coincided with the 2013 reform. Two candidates are worth noting. First, BC experienced a real estate boom after 2013, but this affects construction in structures rather than M&E, which is our outcome variable. Second, differential commodity price movements across provinces could affect investment independently of the PST; the Alberta oil price interaction in Columns (3) and (4) is designed to absorb the most salient of these. The stability of $\hat{\beta}$ across all four specifications suggests these forces do not systematically confound the estimated PST effect.

B Data Appendix

Table B.1: Sector Classification

Sector	Two-digit NAICS Industries
Goods-Producing	Agriculture, forestry, fishing and hunting; Mining, quarrying, and oil and gas extraction; Utilities; Construction; and Manufacturing.
Services	Trade; Transportation and warehousing; Information and cultural industries; Finance and insurance; Real estate and rental and leasing; Professional, scientific and technical services; Management of companies and enterprises; Administrative and support, waste management and remediation services; Educational services; Health care and social assistance; Arts, entertainment and recreation; Accommodation and food services; Public administration; and Other services.
Advanced Manufacturing	Machinery, electrical equipment, transportation equipment, chemicals, plastics, metals, petroleum and coal products, and non-metallic minerals.
Traditional Manufacturing	Food and beverage; textiles; clothing; wood products; paper; printing; furniture; and miscellaneous manufacturing.

Notes: This table reports the sector aggregation used in the analysis, following Statistics Canada's two-digit NAICS classification.

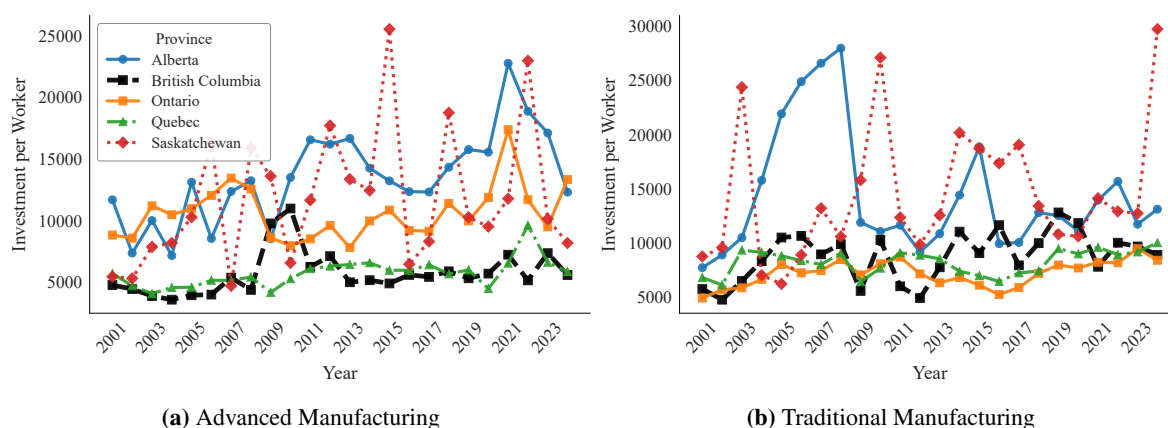


Figure B.1: Manufacturing Investment per Worker: Advanced vs. Traditional, Larger Provinces

Notes: Investment per worker (2017 constant CAD) in advanced and traditional manufacturing (Panel (a) and Panel (b) respectively). BC shown in black. **Sources:** Statistics Canada; authors' calculations.

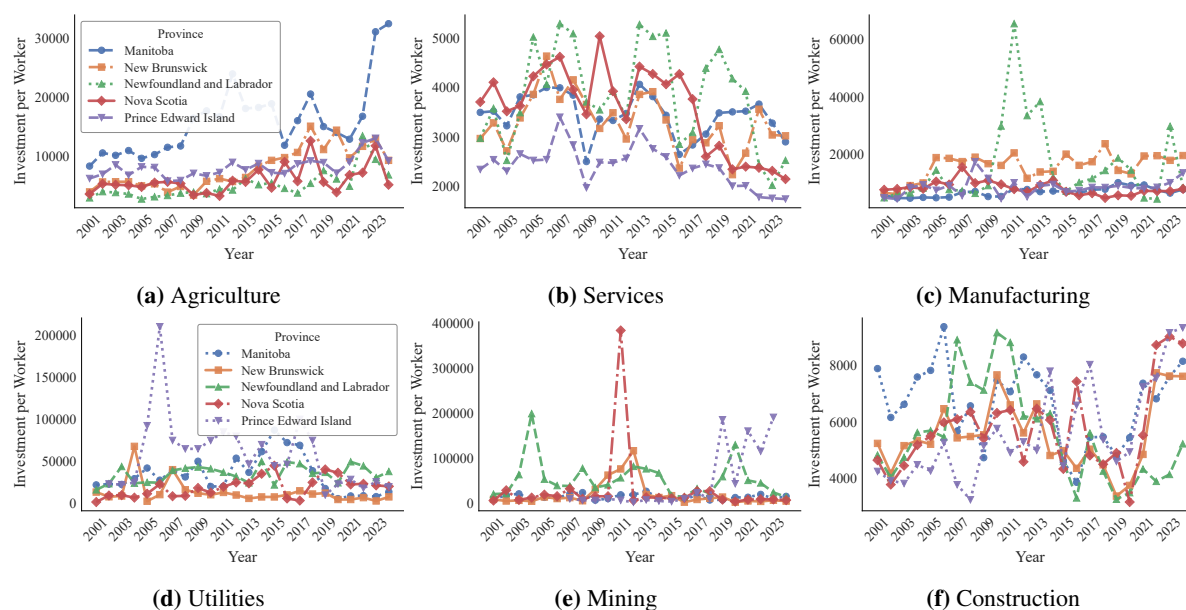


Figure B.2: M&E Investment per Worker by Sector: Other Provinces, 2001–2024

Notes: Investment per worker (2017 constant CAD) for Manitoba, New Brunswick, Newfoundland and Labrador, Nova Scotia, and Prince Edward Island. **Sources:** Statistics Canada; authors' calculations.

C Robustness

A key decision in the baseline estimation is the exclusion of the 2010–2012 HST transition years. The rationale, detailed in Appendix A, is that retaining these years would mix the investment response to the *HST adoption* with the pre-reform baseline, potentially overstating the estimated PST effect. However, this exclusion is itself a specification choice that could be questioned: one might argue that the transition years capture real investment dynamics that inform the counterfactual trend. This appendix directly addresses that concern.

Tables C.2–C.4 reproduce all three main specifications—baseline, sector heterogeneity, and manufacturing—with the 2010–2012 observations reinstated for all provinces. Two results stand out. First, the PST coefficient in the baseline specification falls from -0.063 to a range of -0.049 to -0.052 : somewhat smaller in magnitude, but statistically significant and of the same sign. Second, the sectoral ordering is entirely preserved: Mining and Utilities remain the most adversely affected sectors, Services remain essentially unaffected, and the advanced/traditional differential within manufacturing is maintained. The reduced magnitude is consistent with the interpretation that the HST interlude raised investment above its underlying trend, making the post-2013 contraction appear somewhat larger when the transition years are excluded. Importantly, none of the substantive policy conclusions change under the alternative sample: the evidence for a PST-driven suppression of M&E investment across capital-intensive sectors is robust to this specification choice.

Table C.2: Robustness: Baseline DiD Including Transition Years 2010–2012

Variable	<i>Dependent variable: log(M&E investment per worker)</i>			
	(1)	(2)	(3)	(4)
PST	−0.051** (0.021)	−0.050** (0.022)	−0.052** (0.022)	−0.049** (0.022)
Lagged M&E stock	0.566***	0.599***	0.603***	0.630***
Lagged capital growth	2.555***	2.535***	2.524***	2.454***
Lagged GDP per worker		0.376***	0.380***	0.301**
Alberta × Oil price			−0.0018**	−0.0019**
Lagged employment share				−6.845**
Province FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Observations	3,926	3,921	3,921	3,921
R^2	0.934	0.936	0.936	0.936
Within R^2	0.261	0.279	0.279	0.283

Notes: The 2010–2012 HST transition years are included in the estimation sample for all provinces. The PST indicator equals 1 for BC from 2013 onward and 0 otherwise. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3: Robustness: PST Effects Across Broad Sectors Including Transition Years 2010–2012

Variable	<i>Dependent variable: log(M&E investment per worker)</i>			
	(1)	(2)	(3)	(4)
PST × Agriculture (ref.)	0.528*** (0.040)	0.459*** (0.027)	0.515*** (0.035)	0.377*** (0.042)
PST × Construction	−0.507*** (0.022)	−0.623*** (0.023)	−0.593*** (0.025)	−0.419*** (0.049)
PST × Manufacturing	−0.574*** (0.008)	−0.306*** (0.023)	−0.327*** (0.017)	−0.133*** (0.040)
PST × Mining	−0.970*** (0.035)	−0.934*** (0.032)	−0.988*** (0.034)	−0.875*** (0.066)
PST × Services	−0.554*** (0.031)	−0.478*** (0.028)	−0.496*** (0.028)	−0.346*** (0.036)
PST × Utilities	−0.949*** (0.023)	−0.867*** (0.022)	−0.850*** (0.020)	−0.702*** (0.028)
Lagged M&E growth	2.761***	2.329***	2.234***	2.173***
Lagged M&E intensity		2.625***	2.017***	1.958***
Lagged capital per worker			0.618***	0.670***
Lagged employment share				−0.395***
Province FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Observations	3,926	3,926	3,926	3,921
R^2	0.924	0.932	0.939	0.939
Within R^2	0.157	0.244	0.316	0.323

Notes: Robustness estimates retaining all provinces' observations for 2010–2012. All interaction coefficients are reported relative to Agriculture (reference category). The total PST effect for sector k equals the Agriculture coefficient plus the interaction term for sector k . Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Robustness: PST Effects Within Manufacturing Including Transition Years 2010–2012

Variable	<i>Dependent variable: log(M&E investment per worker)</i>			
	(1)	(2)	(3)	(4)
PST (Advanced Manufacturing, ref.)	0.125** (0.040)	0.111** (0.036)	0.111*** (0.034)	0.095** (0.034)
PST × Traditional Manufacturing	−0.077* (0.042)	−0.064** (0.028)	−0.086*** (0.024)	−0.077** (0.026)
Lagged capital per worker	0.607***	0.673***	0.611***	0.562***
Lagged M&E intensity		1.478***	1.312***	1.339***
Lagged sector GDP			0.105**	0.232**
Lagged employment share				−1.778*
Province FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Observations	2,871	2,871	2,871	2,871
R^2	0.648	0.668	0.672	0.675
Within R^2	0.179	0.227	0.237	0.242

Notes: Robustness estimates retaining all provinces' observations for 2010–2012. Advanced manufacturing is the reference group; the total effect for traditional manufacturing equals the PST coefficient plus the interaction term. Standard errors clustered at the province level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.